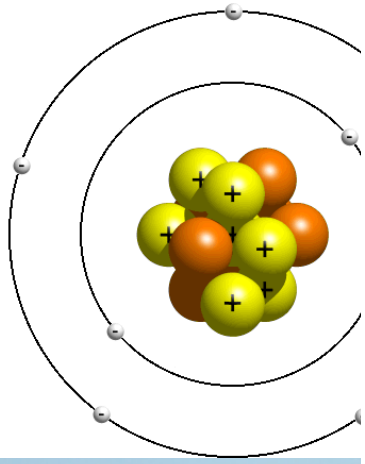
When Physics and Computer Science Collide

A Cross Cultural Extravaganza

Dave Bacon
Department of Computer Science & Engineering
University of Washington

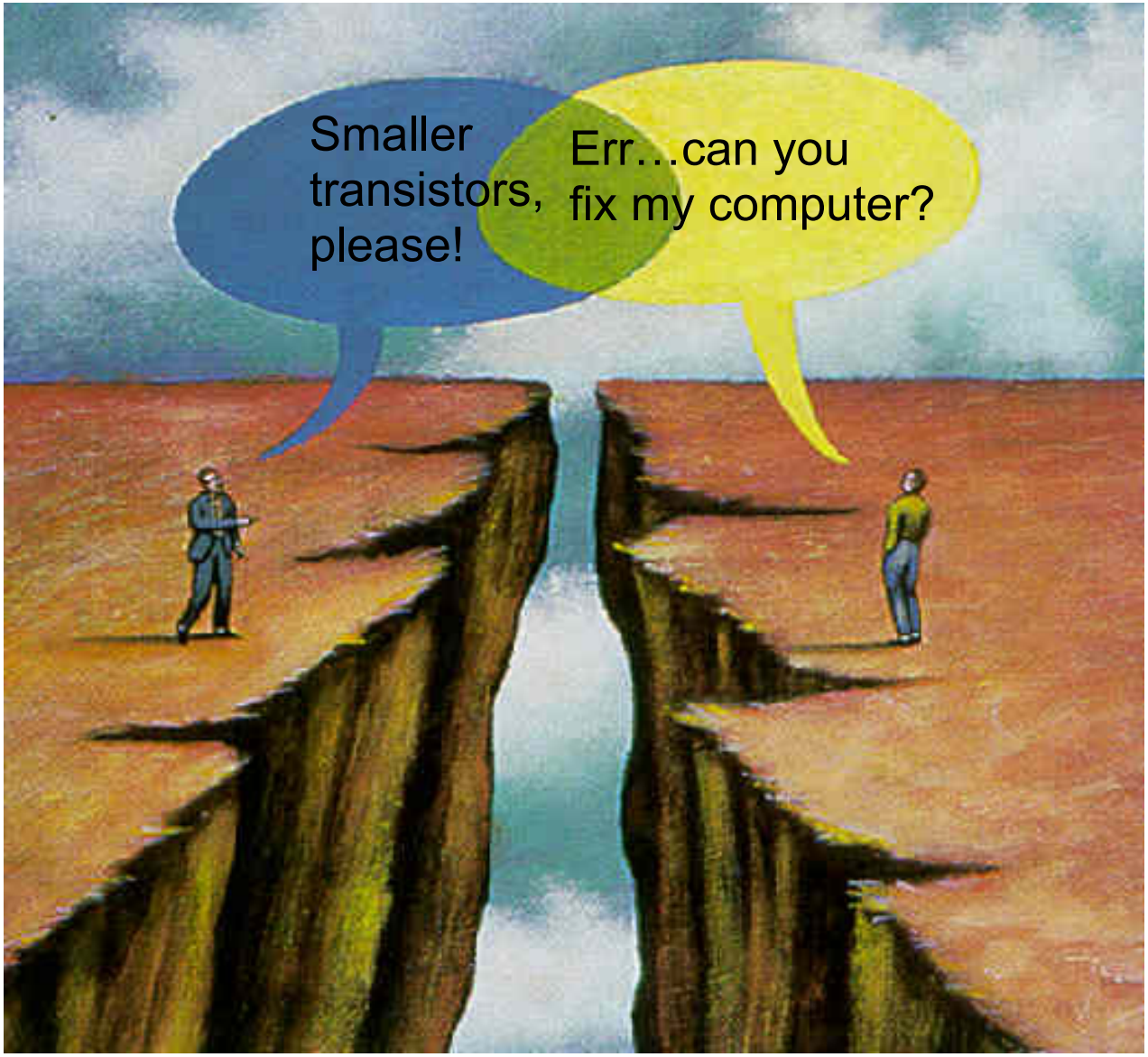











Smaller transistors, please!
Err…can you fix my computer?

Obligatory Feynman Anecdote

The **Danger** of Interdisciplinary Work

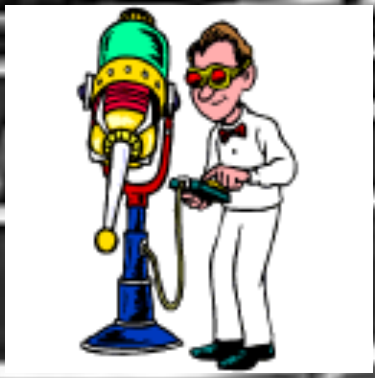
Wow that biophysicist sure knows a lot of physics!



Wow that biophysicist sure knows a lot of biology!



Biologist



Physicis
t

My Goal







Quantum Algorithms

|QW⟩ group

Fault-Tolerant Quantum Computing

Quantum Computer?

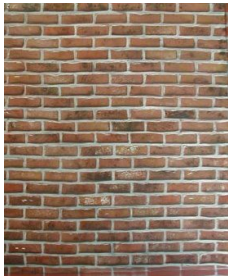




Richard Feynman

"How can we simulate quantum mechanics? ...Can we do it with a new kind of computer… a quantum computer? It's not a Turing machine but a machine of a different kind" - Richard Feynman (1981)





simulatin
g
quantum
theory

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059

=

398075086424064937397125500550386491199064362342526708406385189575946388957261768583317

×

472772146107435302536223071973048224632914695302097116459852171130520711256363590397527

FACTORING

Classical: $O(\exp(n^{1/3}))$

Quantum: $O(n^2 \log n)$

Peter Shor

Quantum Computers!





Public Key Cryptosystems

RSA

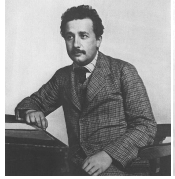
BROKEN

Curves



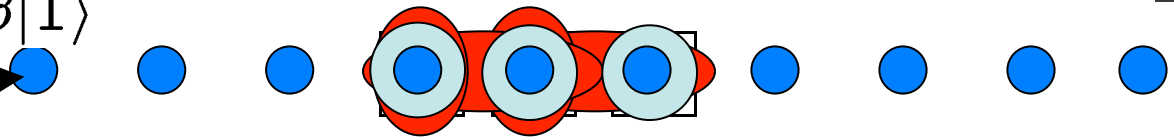
188198812920607963838697239461650439807163563379417382700763356422988859715234654853190606065047430453173880113033967161969232120573403187955065699622130516875930765025705 9

Quantum Computer




$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

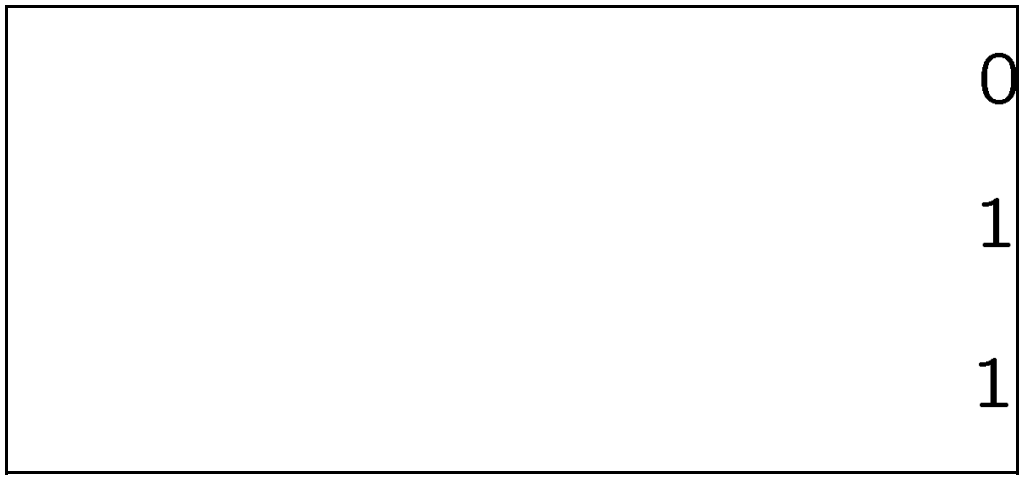
qubits



$$\frac{1}{2}(|000\rangle + |011\rangle + |100\rangle - |111\rangle)$$

Quantum algorithm

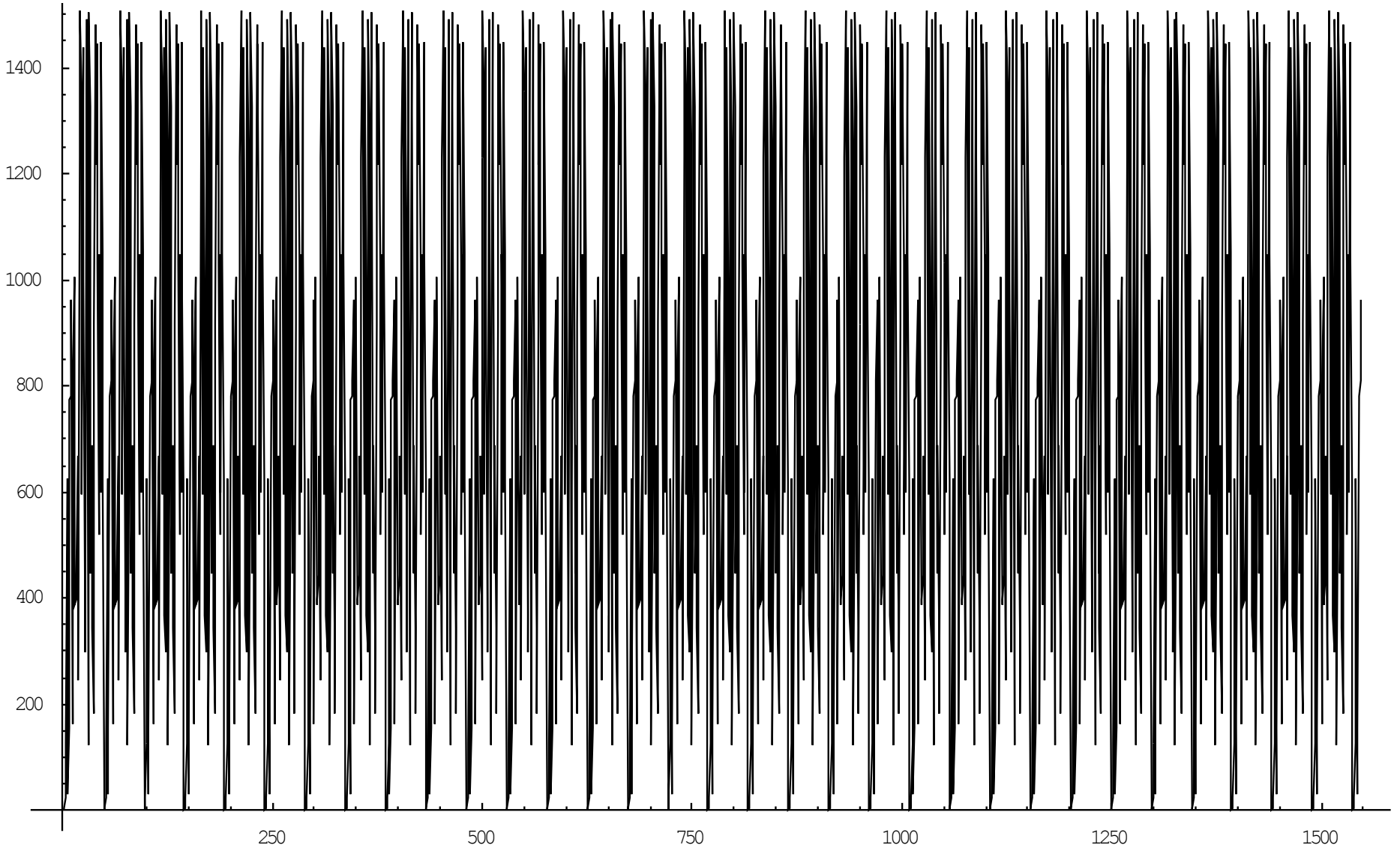
1. Initialize
2. Series of unitary gates
3. Measurement



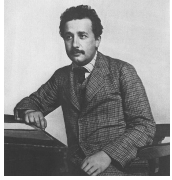
Shor's Algorithm



N=1547, r=5, Period = 48. 48+1=49 GCD(1547,49)=7



Period Finding

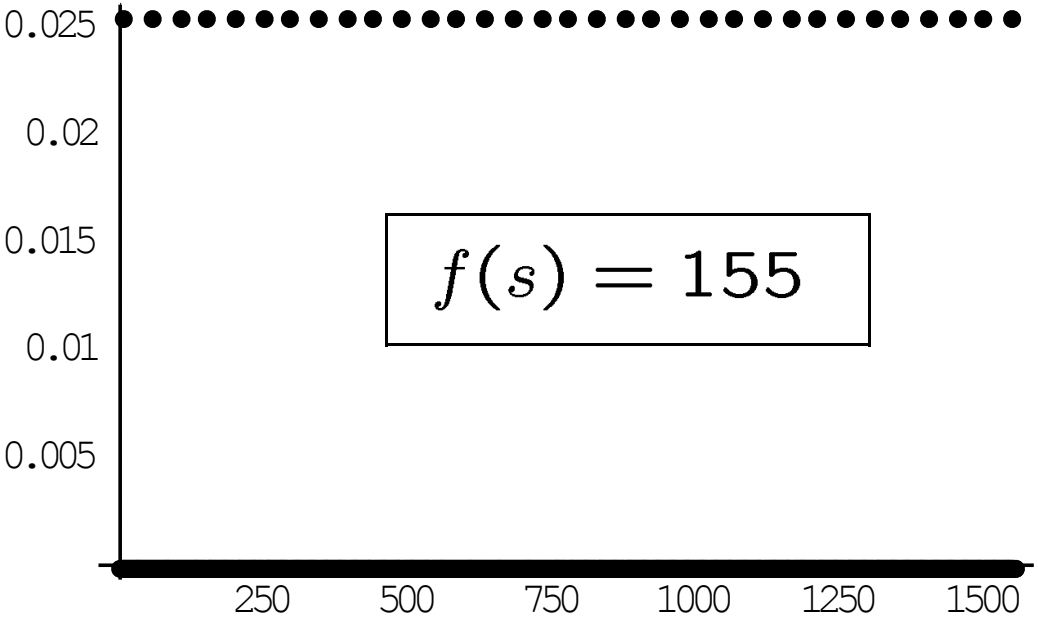


Why hard?

$$f(s) = r^s \bmod N$$



$$\frac{1}{\sqrt{N}} \sum_{s=0}^{N-1} |s\rangle \otimes |f(s)\rangle$$



$$f(s) = 155$$

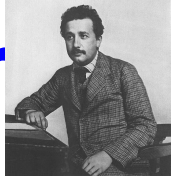
Measure second register

First register is (nearly) symmetric under translation

$$T|x\rangle = |x+1\rangle$$

$$T\rho T^\dagger \approx \rho$$

The First Basis Change We Learn In Quantum Mechanics

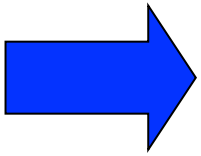


$$\frac{1}{\sqrt{N}} \sum_{s=0}^{N-1} |s\rangle \otimes |f(s)\rangle$$

symmetry under translation

$$T|x\rangle = |x+1\rangle$$

$$T\rho T^{\dagger} = \rho$$



change to momentum basis!

$$|p\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} \omega_N^{ixp} |x\rangle$$

$$\omega_N = \exp\left(\frac{2\pi i}{N}\right)$$

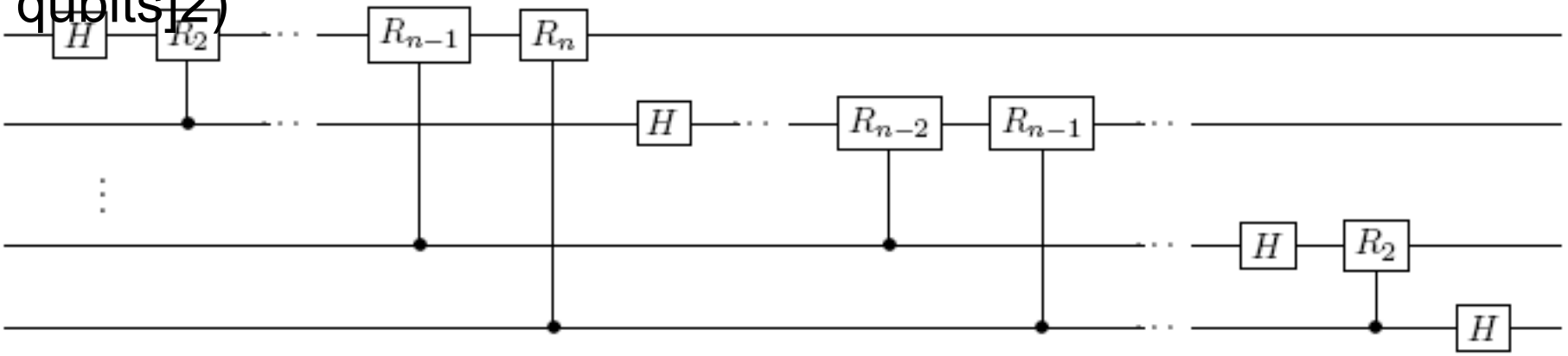
Perform a Fourier Transform!

Quantum Fourier Transform



Efficient quantum circuit for quantum Fourier transform:

[Efficient = O([# qubits]2)



Factoring



Period Finding

Symmetry





Change Basis

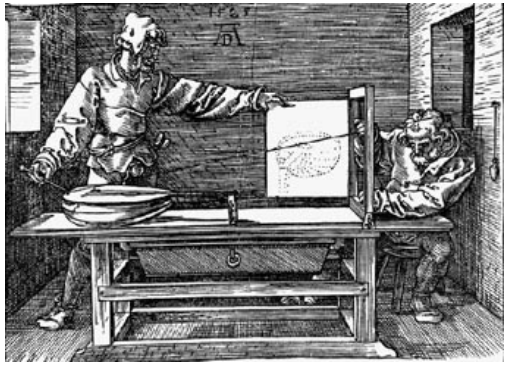
Efficient Circuit



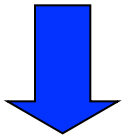


Shor's Algorithm

Moral

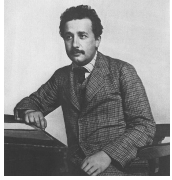


Symmetries



Quantum algorithms

The Second Basis Change We Learn In Quantum Mechanics?



The Second Basis Change We Learn In Quantum Mechanics:

Clebsch-Gordan Transform

Singlet:

$$|S=0,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle - |\downarrow\rangle \otimes |\uparrow\rangle)$$

Triplet:

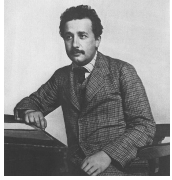
$$|S=1,m=1\rangle = |\uparrow\rangle \otimes |\uparrow\rangle$$

$$|S=1,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle + |\downarrow\rangle \otimes |\uparrow\rangle)$$

$$|S=1,m=-1\rangle = |\downarrow\rangle \otimes |\downarrow\rangle$$

What Use?

Collective Unitaries



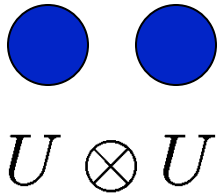
$$|S=0,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle - |\downarrow\rangle \otimes |\uparrow\rangle)$$

$$|S=1,m=1\rangle = |\uparrow\rangle \otimes |\uparrow\rangle$$

$$|S=1,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle + |\downarrow\rangle \otimes |\uparrow\rangle)$$

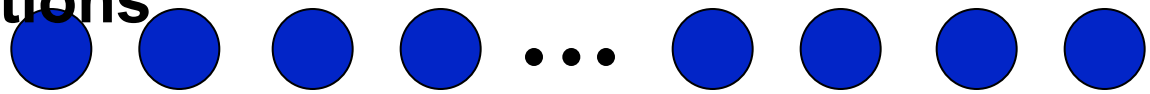
$$|S=1,m=-1\rangle = |\downarrow\rangle \otimes |\downarrow\rangle$$

$$\left[\begin{array}{c|c} 1 & 0 \\ \hline 0 & U \end{array}\right]$$



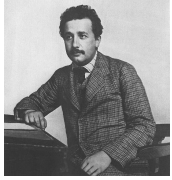
$U \otimes U$

Symmetry Under n Collective Unitary Rotations



$U^{\otimes n}$

The Symmetric Group



Transform Under SWAP

Singlet:

$$|S=0,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle - |\downarrow\rangle \otimes |\uparrow\rangle) \qquad -1$$

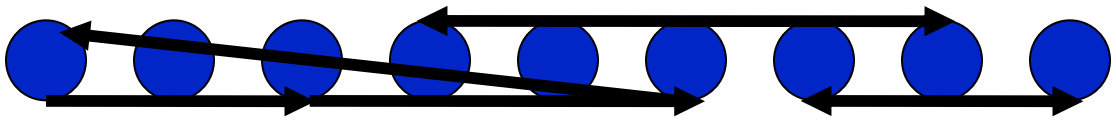
Triplet:

$$|S=1,m=1\rangle = |\uparrow\rangle \otimes |\uparrow\rangle$$

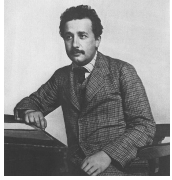
$$|S=1,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle + |\downarrow\rangle \otimes |\uparrow\rangle) \qquad +1$$

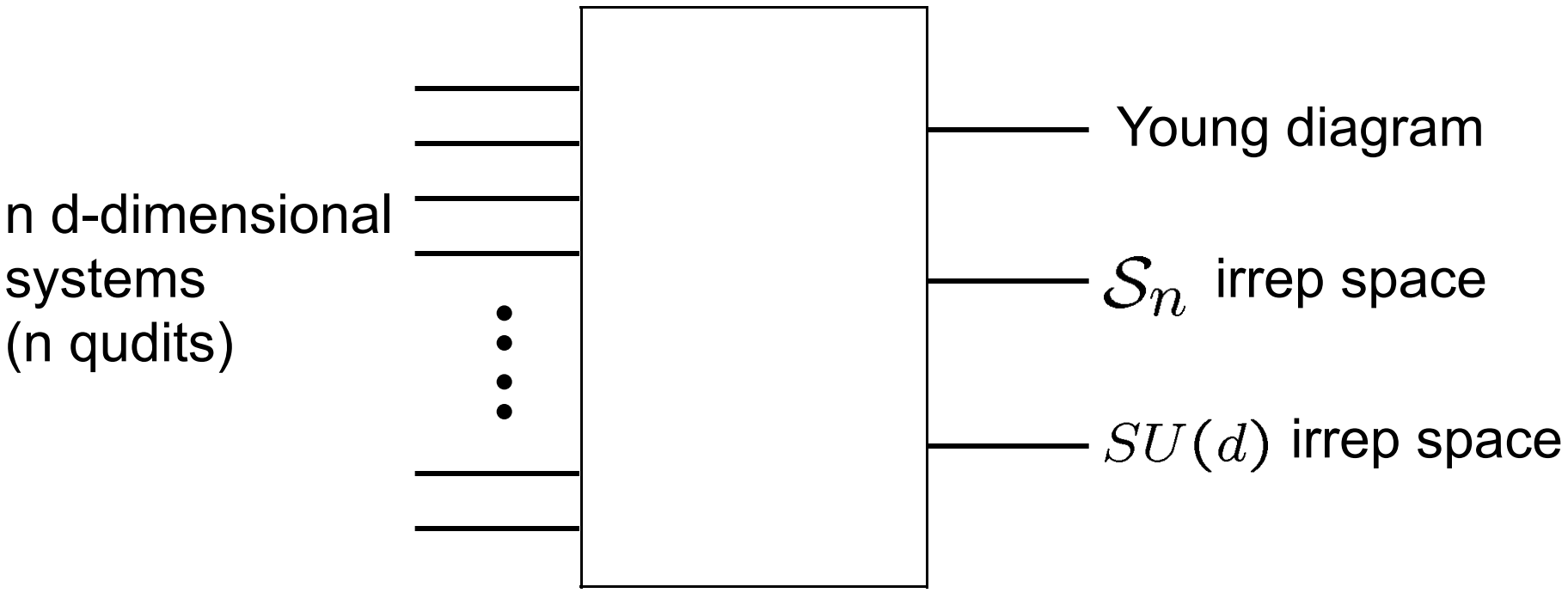
$$|S=1,m=-1\rangle = |\downarrow\rangle \otimes |\downarrow\rangle$$

Symmetry Under Permutation of n Subsystems



Quantum Schur Transform

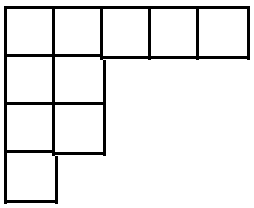




Schur duality

$$[\mathbb{C}^d]^{\otimes n} \overbrace{\cong}^{\text{Schur}} \bigoplus_{\mathcal{Y}} S_{\mathcal{Y}} \otimes U_{\mathcal{Y}}$$

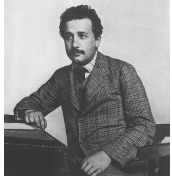
Young diagram

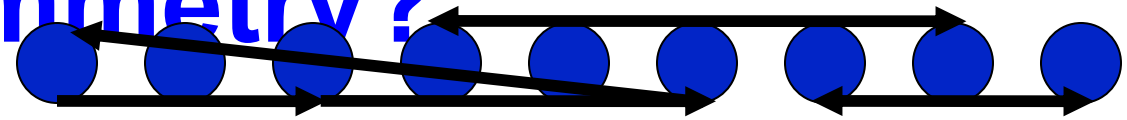


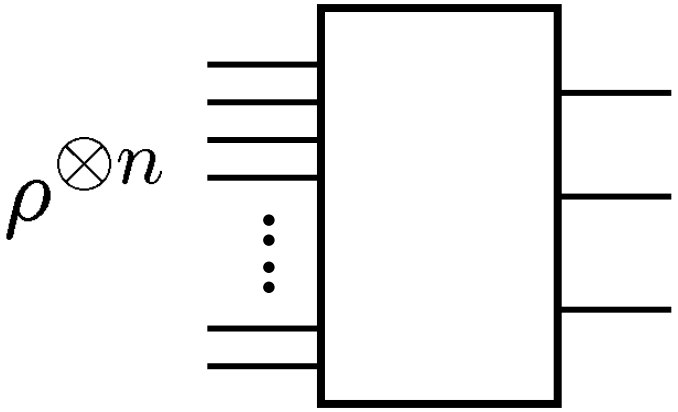
$$k_1 \geq k_2 \geq \cdots \geq k_d$$

$$\sum_{i=1}^{d} k_i = n$$

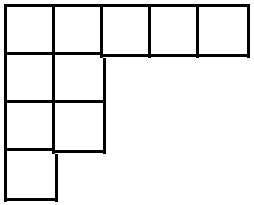
What Use, This Symmetry?







Young diagram



$$k_1 \geq k_2 \geq \cdots \geq k_d$$

$$\sum_{i=1}^{d} k_i = n$$

$$\rho = \sum_{i=1}^{d} p_i |\phi_i\rangle\langle\phi_i|$$

Spectrum

$$\left(\frac{k_1}{n}, \frac{k_2}{n}, \ldots, \frac{k_d}{n}\right)$$

"Optimal" estimate of spectrum

[M. Keyl and R. F. Werner. Phys. Rev. A, 2001]

The Plethora of Uses of Schur



Optimal spectrum estimation

[M. Keyl and R. F. Werner. Phys. Rev. A, 2001]

Universal distortion-free entanglement concentration

[M. Hayashi and K. Matsumoto. Information Theory, 2004]

Universal compression with optimal overflow exponent

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[M. Hayashi, J. Phys. A, 2002]

Communicating without a shared reference frame

[S. Bartlett, T. Rudolph, and R. Spekkens, Phys. Rev. Lett., 2003]

Encoding into decoherence-free (noiseless) subsystems

[J. Kempe, D. Bacon, D. A. Lidar, and K. B. Whaley, Phys. Rev. A, 2001]

Simulating lattice gauge theories on a quantum computer

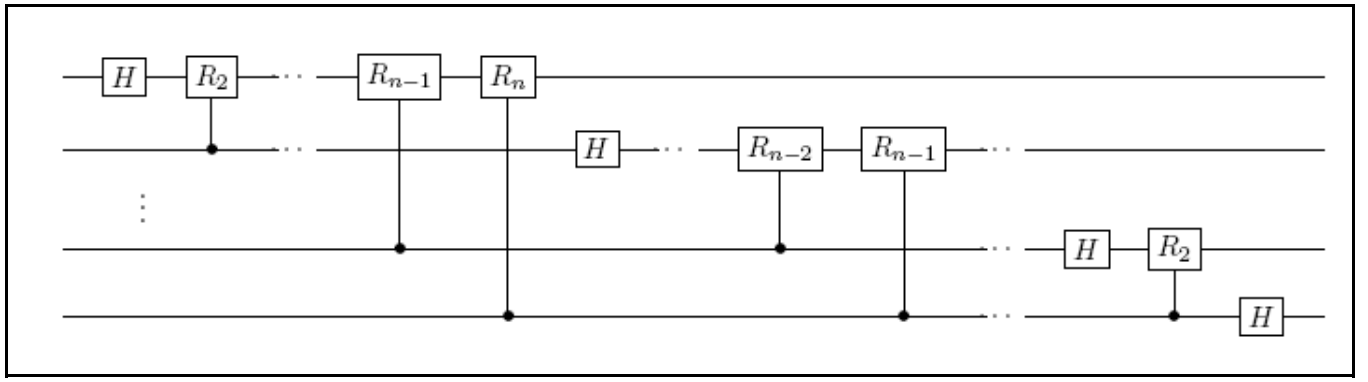
[T. Byrnes and Y. Yoshihisa, Phys. Rev. A 2006]

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But Efficient?



QFT



Schu
r

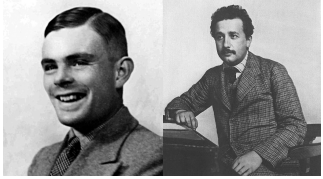
?

Who would need to efficiently calculate Clebsch-Gordan coefficients for SU(d)?



[L. C. Biedenharn and J. D. Louck, Commun. Math. Phys. 1968]

Efficient Schur



[D. Bacon, I.L. Chuang, A. Harrow, Phys. Rev. Lett. 2006]
[D. Bacon, I.L. Chuang, A. Harrow, SODA 2007]

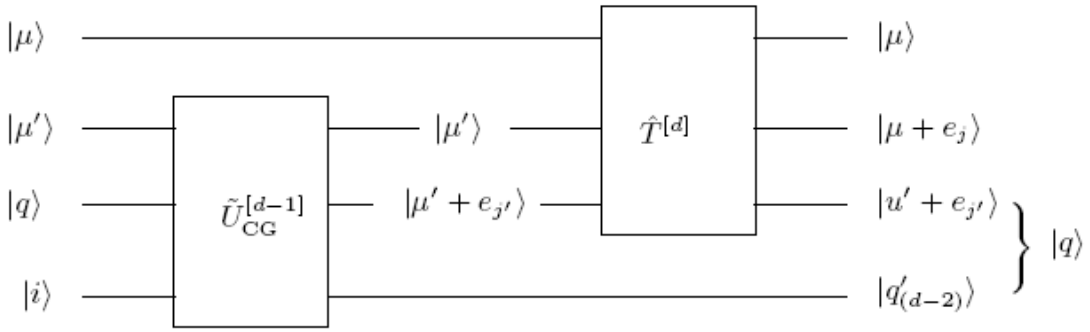


FIG. 4: The $\mathcal{U}_d$ CG transform, $\mathbf{U}_{\mathrm{CG}}^{[d]}$, is decomposed into a $\mathcal{U}_{d-1}$ CG transform $\tilde{U}_{\mathrm{CG}}^{[d-1]}$ (see Eq. (54)) and a reduced Wigner operator $\hat{T}^{[d]}$. In Fig. 5 we show how to reduce the reduced Wigner operator to a $d \times d$ matrix conditioned on μ and $\mu' + e_{j'}$.

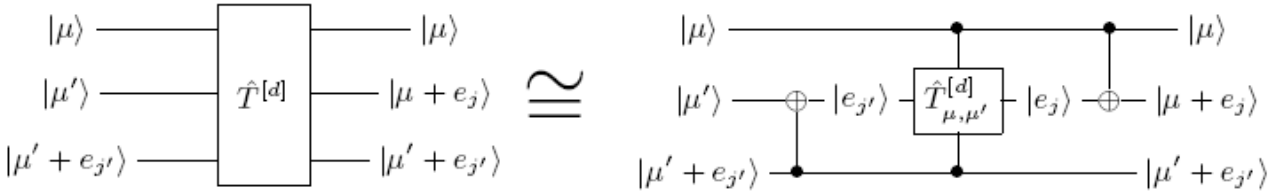


FIG. 5: The reduced Wigner transform $\hat{T}^{[d]}$ can be expressed as a $d \times d$ rotation whose coefficients are controlled by μ and $\mu' + e_{j'}$.

Schur transform to accuracy ϵ : time $poly\left(log\frac{1}{\epsilon}, n, d\right)$

The Plethora of Uses of Schur



Optimal spectrum estimation

[M. Keyl and R. F. Werner. Phys. Rev. A, 2001]

Universal distortion-free entanglement concentration

[M. Hayashi and K. Matsumoto. Information Theory, 2004]

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[T. Byrnes and Y. Yoshihisa, Phys. Rev. A 2006]

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But Algorithms?

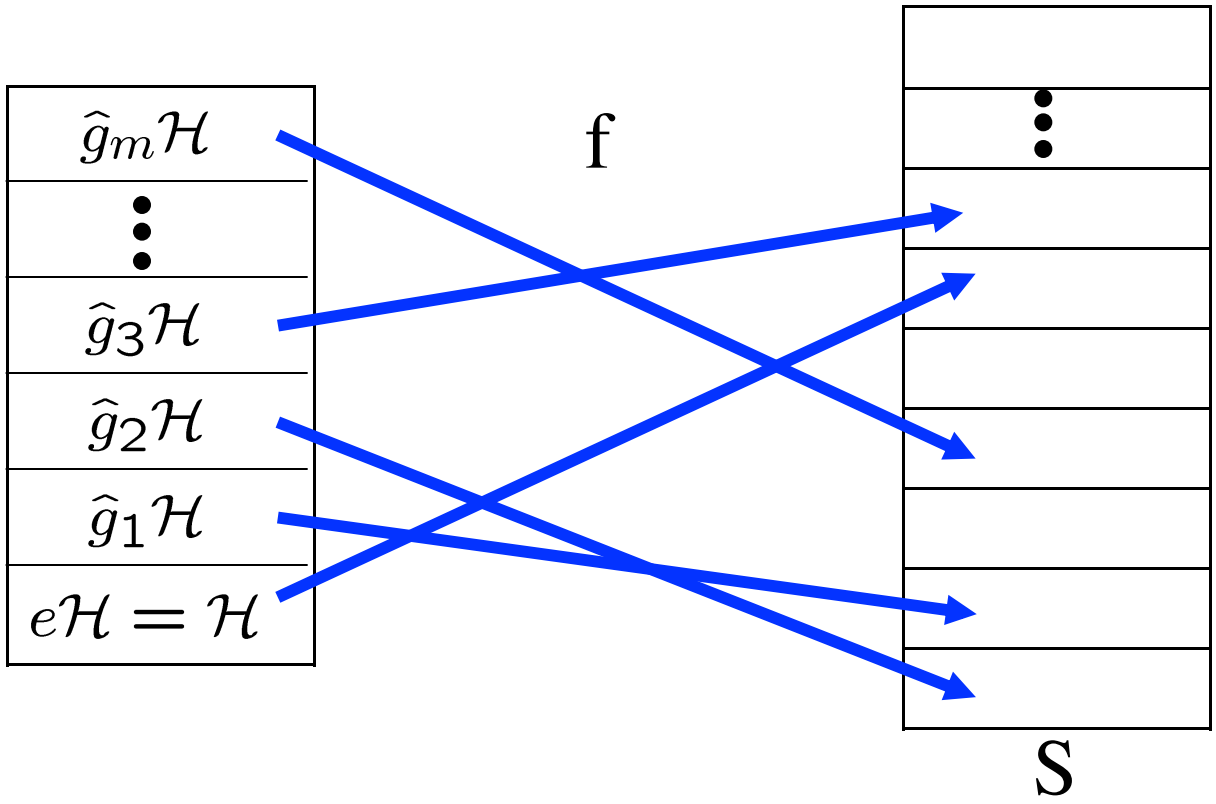
The Hidden Subgroup Problem

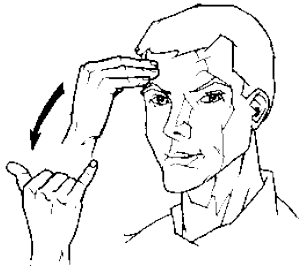


Setup: Fix a group $\mathcal{G}$ (known) and a subgroup $\mathcal{H} \subset \mathcal{G}$ (unknown). Given query access to a function $f : \mathcal{G} \to S$ that is constant and distinct on left (right) cosets of $\mathcal{H}$ in $\mathcal{G}$. We say that f hides $\mathcal{H}$.

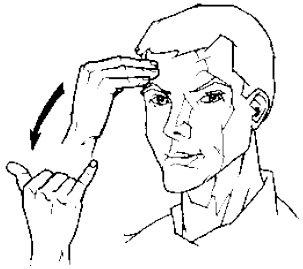
Problem: Find a generating set for $\mathcal{H}$ ("identify $\mathcal{H}$").

Efficient: If runs in time polylog($|\mathcal{G}|$)





Why the Fuss?





Solved Problems (Shor 1994)

efficient algorithm for HSP over $\mathbb{Z}_N$ → efficient algorithm for factoring whole numbers

efficient algorithm for HSP over $\mathbb{Z}_N \times \mathbb{Z}_N$ → efficient algorithm for the discrete logarithm problem

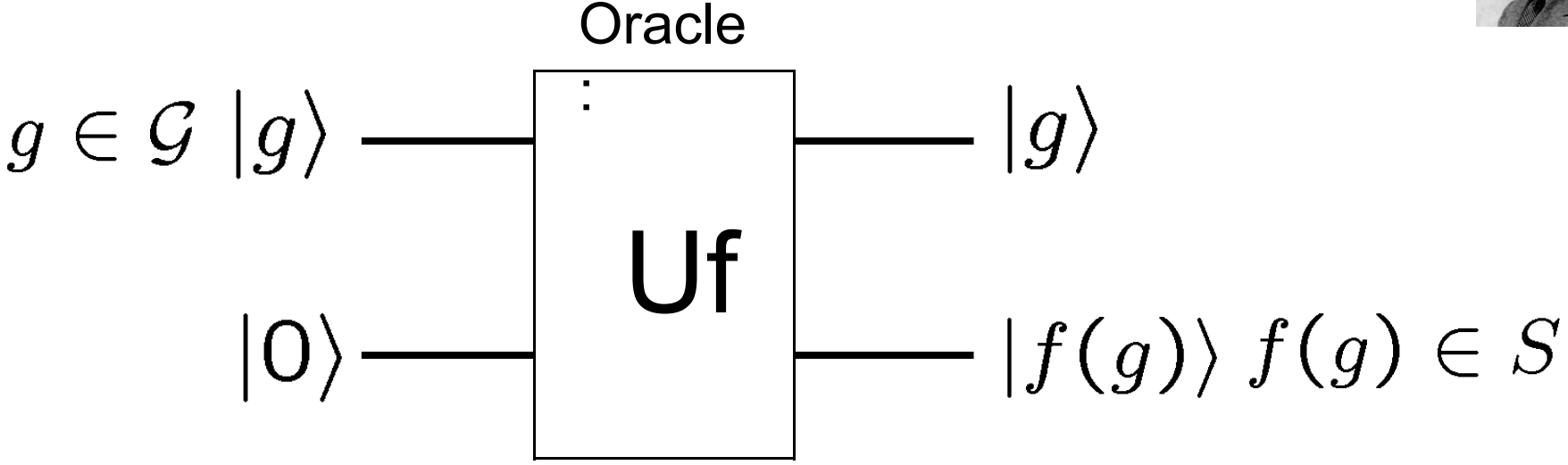
Open Problems

efficient algorithm for HSP over $\mathcal{S}_n \wr \mathbb{Z}_2$ (or $\mathcal{S}_{2n}$) → efficient algorithm for the graph isomorphism problem

efficient "standard" quantum algorithm for HSP over dihedral group $\mathcal{D}_N$ → efficient algorithm for the poly(*n*) unique shortest vector in a lattice problem

Standard Query Model





Query in equal superposition over group elements

$$\frac{1}{\sqrt{|\mathcal{G}|}} \sum_{g \in \mathcal{G}} |g\rangle|0\rangle \rightarrow \sum_{g \in \mathcal{G}} \frac{1}{\sqrt{|\mathcal{G}|}} |g\rangle|f(g)\rangle$$

$$\rho_{\mathcal{H}} = \frac{|\mathcal{H}|}{|\mathcal{G}|} \sum_{g \in \hat{\mathcal{G}}} |g\mathcal{H}\rangle\langle g\mathcal{H}|$$ hidden subgroup state

You Down With Symmetry?

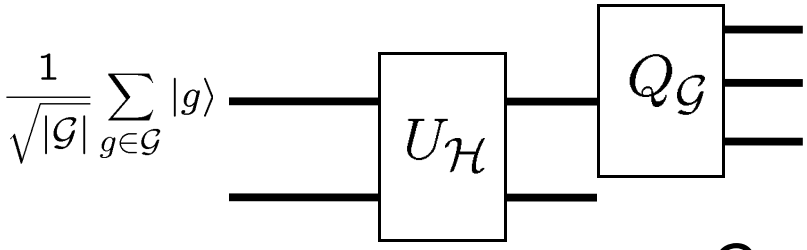


$$\rho_{\mathcal{H}} = \frac{|\mathcal{H}|}{|\mathcal{G}|} \sum_{g \in \hat{\mathcal{G}}} |g\mathcal{H}\rangle\langle g\mathcal{H}| \quad \text{hidden subgroup state}$$

Symmetry under left regular representation:

$$D_L(g)|g'\rangle = |gg'\rangle \qquad D_L(g)\rho_{\mathcal{H}} D_L(g^{-1}) = \rho_{\mathcal{H}}$$

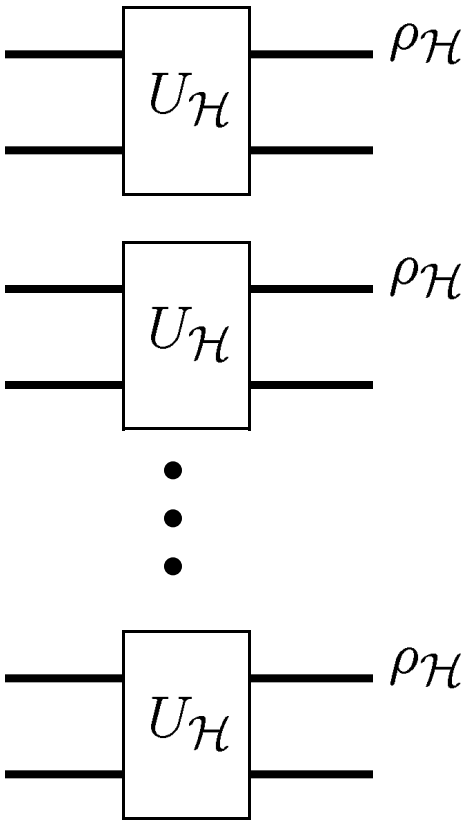
Fourier transform over finite group $\mathcal{G}$



Can we solve HSP this way?

Multicopy





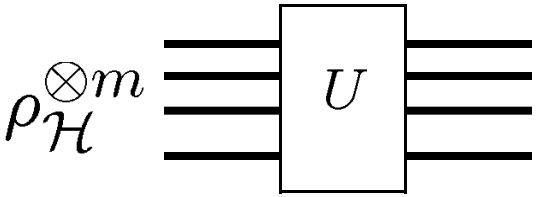
$\rho_{\mathcal{H}}^{\otimes m}$

Can we make a measurement on $m < O(\log |\mathcal{G}|)$ copies and learn the subgroup?

In general (and for important case of symmetric group) **NO**

[Moore, Russell, Hallgren, Roetteler, Sen 05]

Required:



Clebsch-Gordan Transform To The Rescue?



Problem:

$\rho_{\mathcal{H}} \otimes \rho_{\mathcal{H}}$ does not have a Clebsch-Gordan related symmetry

Heisenberg Group $\mathbb{Z}_p^2 \rtimes \mathbb{Z}_p$



Efficient algorithm for the Heisenberg HSP:

[D. Bacon, A.M. Childs, and W. van Dam FOCS 2005]

But Why Does It Work?

Clebsch-Gordan Transform To The Rescue!



Bring the Symmetry to the Problem

Hidden Subgroup Problem

Hidden Subgroup Conjugacy Problem

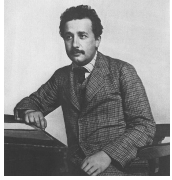
Efficient Heisenberg HSCP Algorithm using CG Transform:

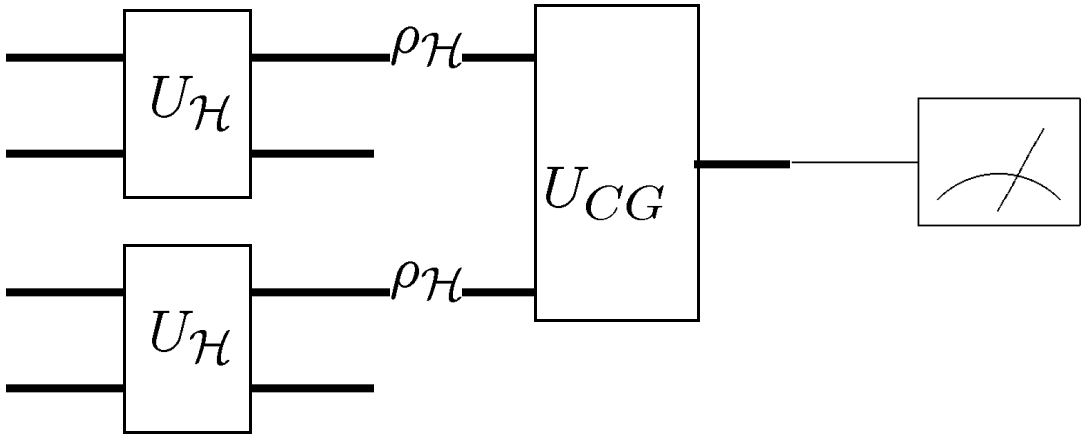
[D. Bacon, quant-ph/0612107, 2006]

Hesienberg HSCP = Heisenberg HSP

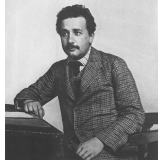
Efficient Clebsch-Gordan Transform for Heisenberg Group

Entangled Measurement





Quantifying Entanglement

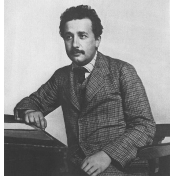


Pure state bipartite entanglement

$$\left[\sum_{i=0}^{r-1}\sqrt{p_i}|\phi_i\rangle\otimes|\varphi_i\rangle\right]^{\otimes m} \longleftrightarrow \left[\frac{1}{\sqrt{2}}(|0\rangle\otimes|0\rangle+|1\rangle\otimes|1\rangle)\right]^{\otimes n}$$

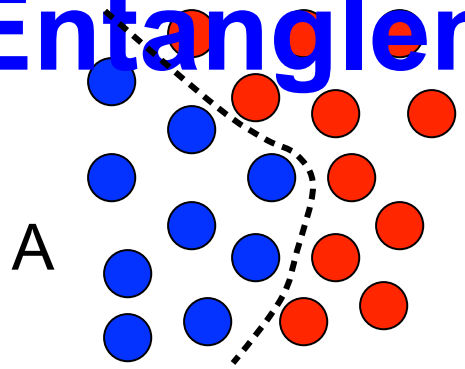
$$\frac{n}{m}=S=-\sum_{i=0}^{r-1}p_i\log_2 p_i$$

Entropy of one-half of mixed state is measure of entanglement



"Small" Entanglement

n qubits



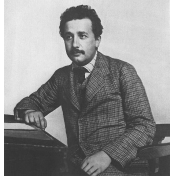
$$\left[\sum_{i=0}^{r-1} \sqrt{p_i}|\phi_i\rangle_A \otimes |\varphi_i\rangle_B\right]^{\otimes m}$$

$$\chi = \max_A S(A) \qquad S(A) = -\sum_{i=0}^{r-1} p_i \log_2 p_i$$

Theorem [Vidal 03]: If χ is *O(poly(n))* then there is an efficient description of the state. If χ remains small throughout a quantum computation then there is a *O(poly(n))* classical simulation of the quantum computation.

"Small" Entanglement = Efficiently Classical Simulation

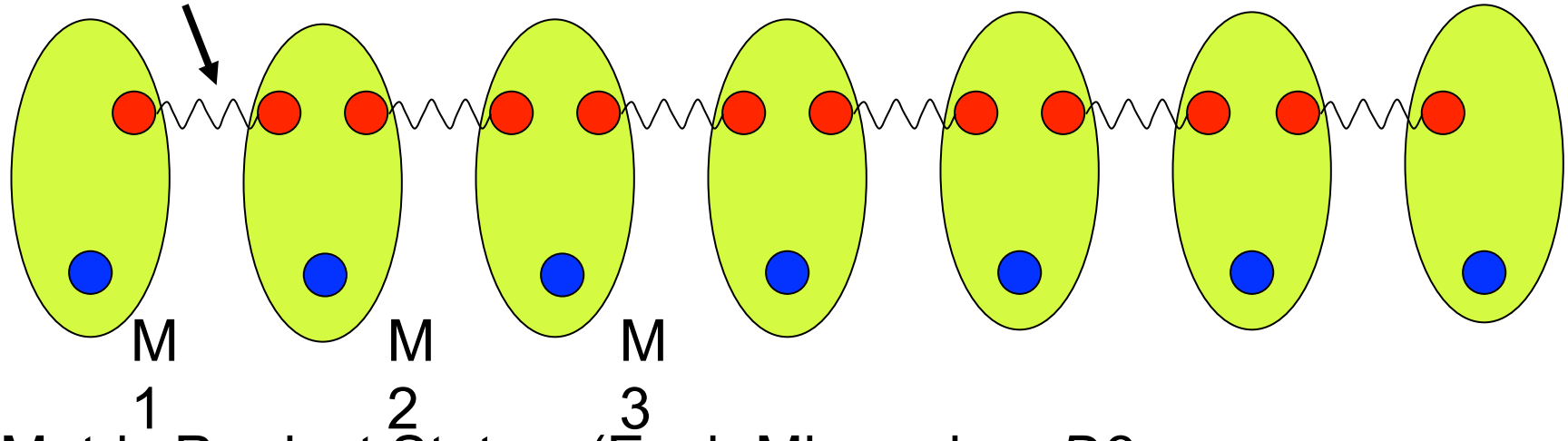
Matrix Product States



"Small" Entanglement = Efficiently classical simulation

? Better classical algorithms for simulating real world quantum systems?

$$\frac{1}{\sqrt{D}} \sum_{i=0}^{D-1} |\phi_i\rangle \otimes |\varphi_i\rangle$$

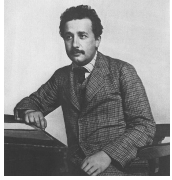


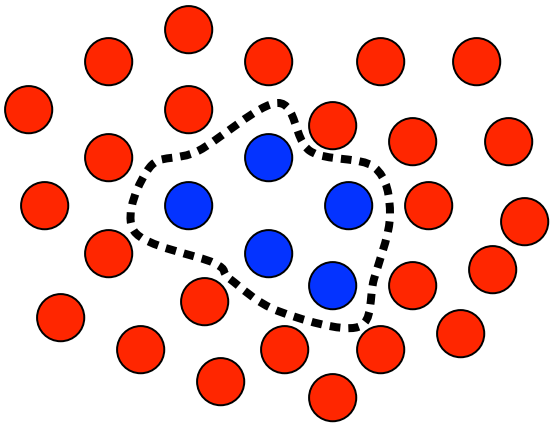
Matrix Product States: (Each Mi requires *D2s* numbers)

Schmidt rank bounded by *D* ⇒ D small, efficient simulation

Density Matrix Renormalization Group [White, 1991]

Perimeter "Law"

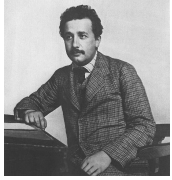


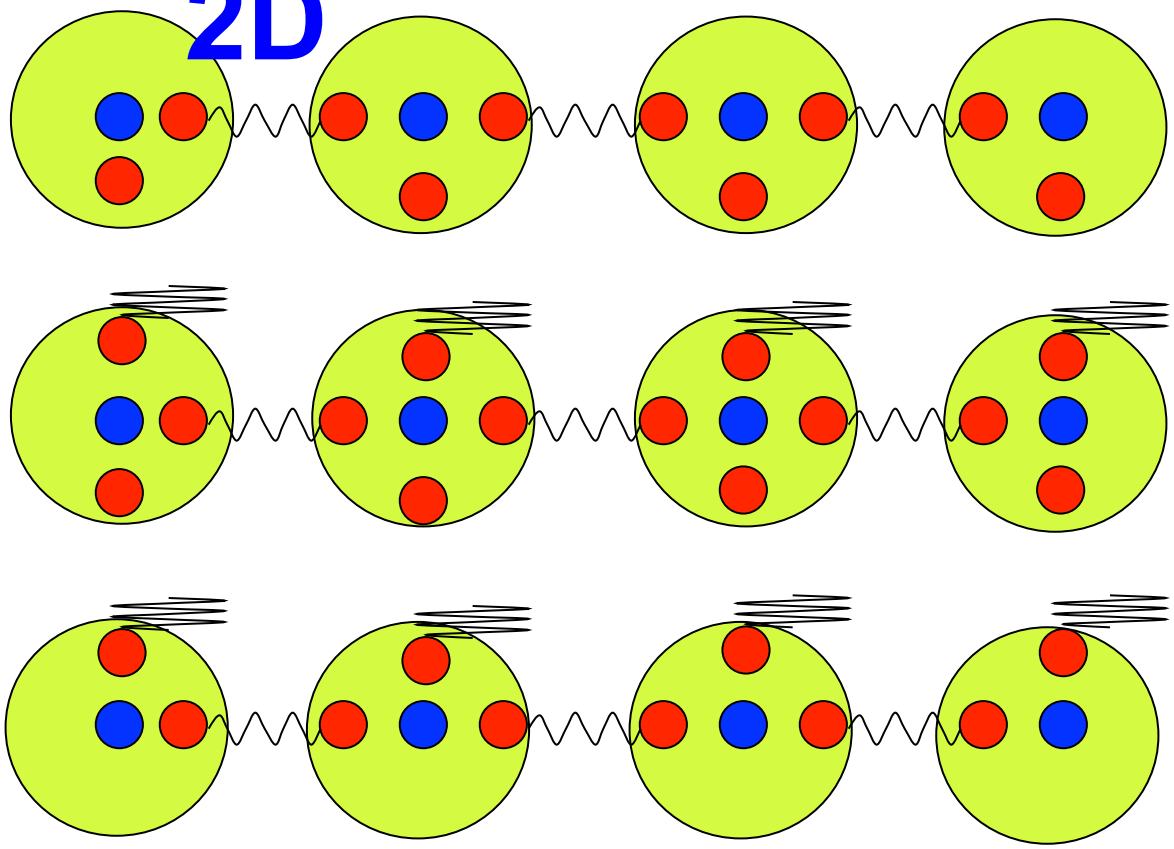


In "many" quantum systems, the low energy sector has an entropy of a region which grows like the perimeter of the region

"Small" entanglement

DMRGish Algorithm in 2D

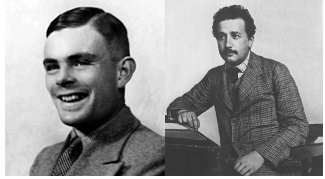




Design states which respect this scaling!

Efficient algorithms for simulating low energy sector of high (spatial) dimensional systems. [Verstraete, Cirac 04]

DMRGish Algorithm

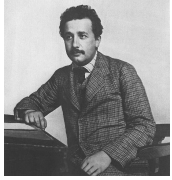




- simulation code (very general)
- efficient new caching data structure
- analytical uses of PEPS states

Gregory Crosswhite
UW Physics Grad Student

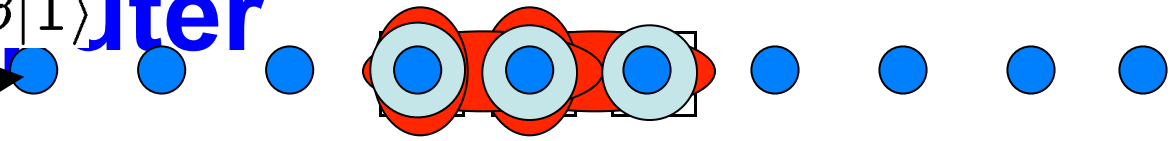
But **what** to simulate with these new algorithms?



The Ideal Quantum Computer

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

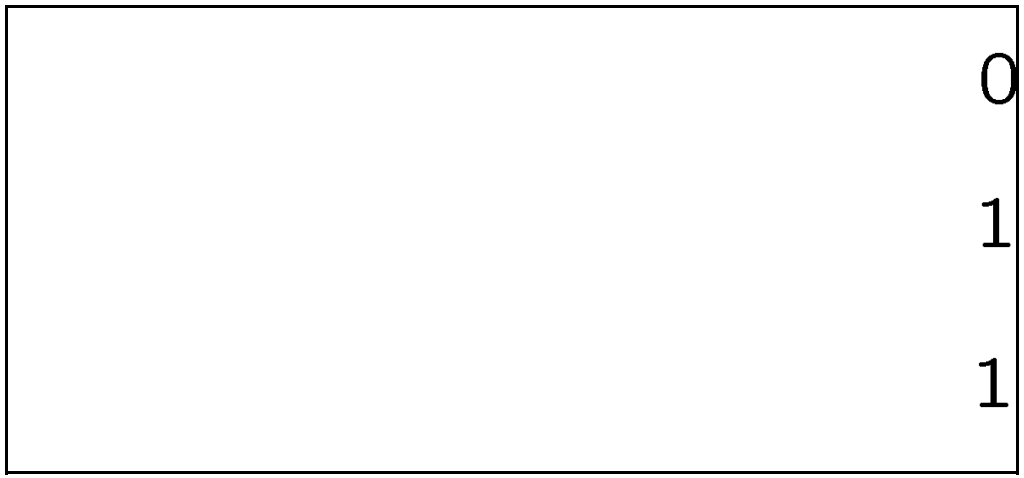
qubits



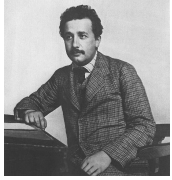
$$\frac{1}{2}(|000\rangle + |011\rangle + |100\rangle - |111\rangle)$$

Quantum algorithm

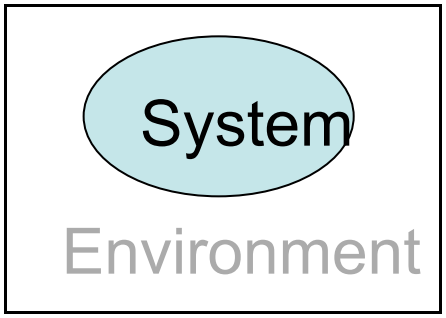
1. Initialize
2. Series of unitary gates
3. Measurement



Obstructions



DECOHERENCE



Total Hamiltonian

$$H = H_S \otimes I + I \otimes H_E + H_I$$

system environmen interaction

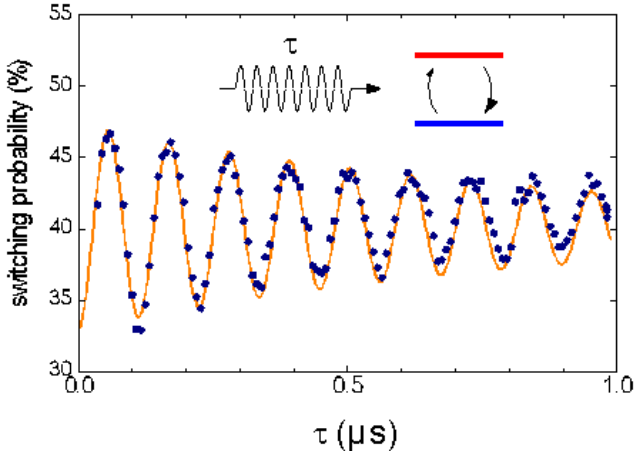
open quantum systems can become *entangled* with their environment

$$(\alpha|0\rangle_S + \beta|1\rangle_S) \otimes |0\rangle_E \rightarrow \alpha|0\rangle_S \otimes |0\rangle_E + \beta|1\rangle_S \otimes |1\rangle_E$$

coherent superposition

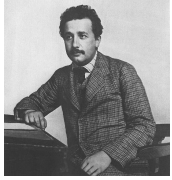
incoherent sum

$$(\alpha|0\rangle_S + \beta|1\rangle_S) \rightarrow \rho = |\alpha|^2 |0\rangle_S\langle 0|_S + |\beta|^2 |1\rangle_S\langle 1|_S$$

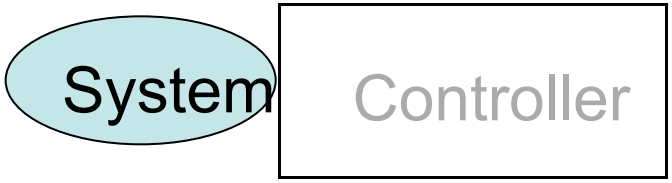


Quantronium (CEA, France)

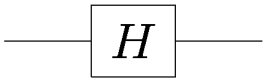
More Obstructions

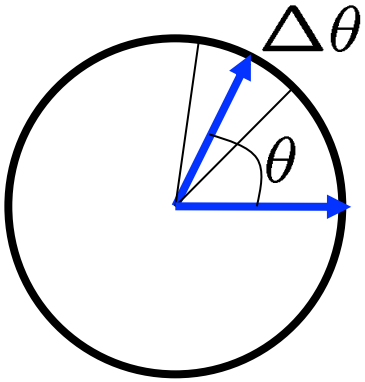


FAULTY CONTROL

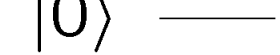


Imprecise or faulty unitary gates



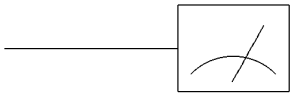


Imperfect state preparation



$$\rho \rightarrow (1-\epsilon)|0\rangle\langle 0| + \epsilon|1\rangle\langle 1|$$

Imperfect measurements

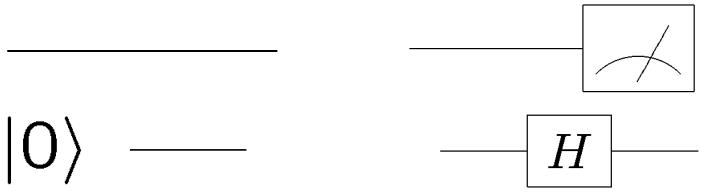


$$\alpha|0\rangle + \beta|1\rangle \begin{cases} Pr(0) = |\alpha|^2 - \epsilon \\ Pr(1) = |\beta|^2 + \epsilon \end{cases}$$

This is the End, My Only Friend, the End



Everything, (quantum) noisy, all the time.



Is a Quantum Computer Even a Computer?

A skeptics checklist:

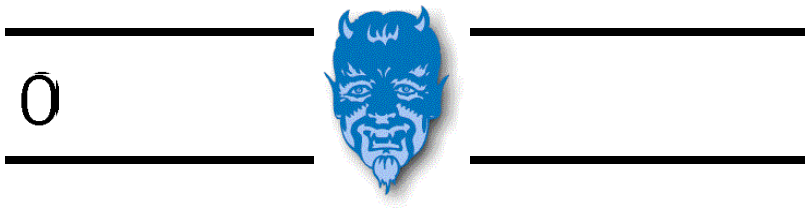


Laughlin

1. quantum data is analog
2. quantum data likes to become classical data
3. quantum evolution is analog
4. quantum theory is wrong

Classical Solutions

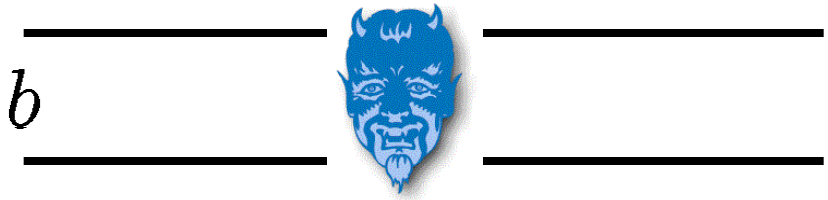




Probability of bit flip = *p*

Solution: redundancy

$0 \rightarrow 000$
$1 \rightarrow 111$



$$\left.\begin{matrix}000\\001\\010\\100\end{matrix}\right\} \rightarrow 0$$

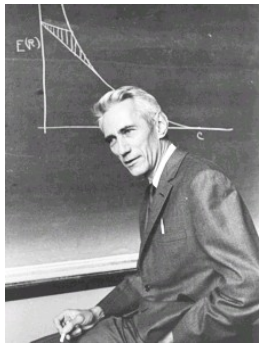
$$\left.\begin{matrix}111\\110\\101\\011\end{matrix}\right\} \rightarrow 1$$

Probability of failure =*3(1-p)p2+p3*

=3p2-2p3

<p if *p<1/2*

Use redundancy to protect information against noise

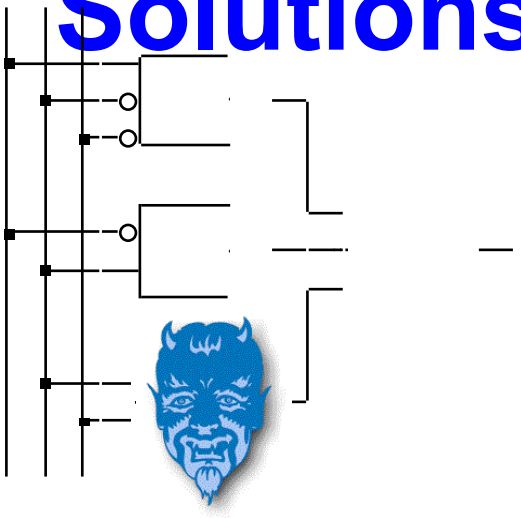


Claude Shannon 1948

“A mathematical theory of communication”

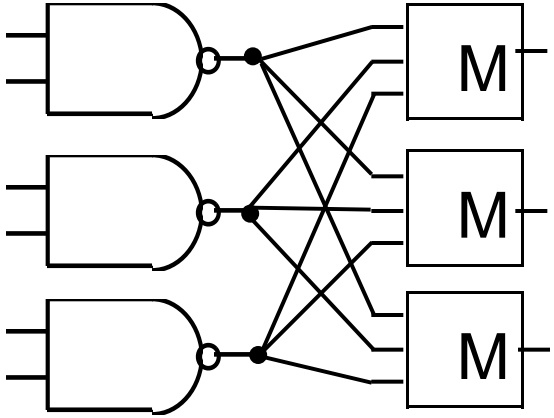
More Classical Solutions





Classical gates fail with probability p

Solution: error correcting codes + fault tolerant construction



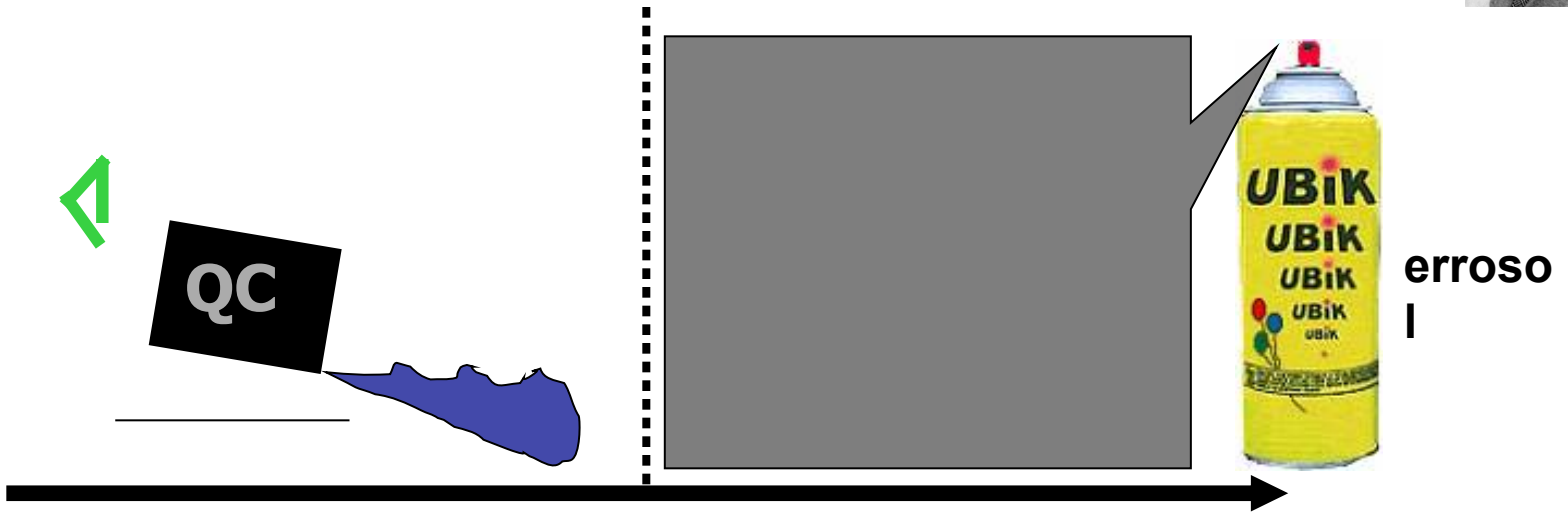


John von Neumann, 1956

"Probabilistic logics and the synthesis of reliable organisms from unreliable components"

Quantum Threshold Theorem



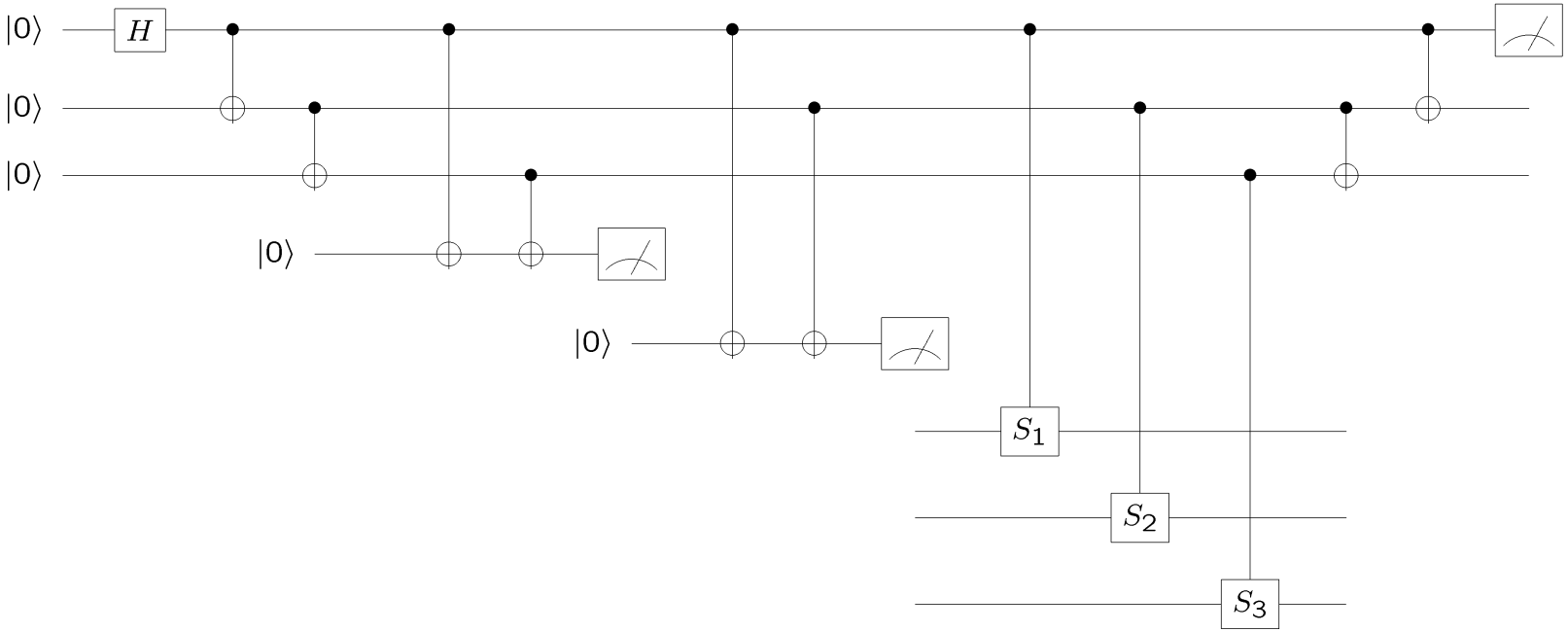


Threshold Theorem for Fault-Tolerant Quantum Computation

[Aharonov, Ben-Or, Knill, LaFlamme, Zurek,
Kitaev, Gottesman, Preskill, Shor, + others 1995-1996]

If noise is weak enough and control is precise enough, then a quantum computation to a desired accuracy can be carried out with an overhead polynomial in the log of the accuracy.

Fault-Tolerant Quantum Computation

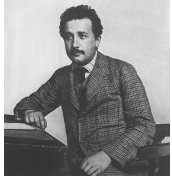


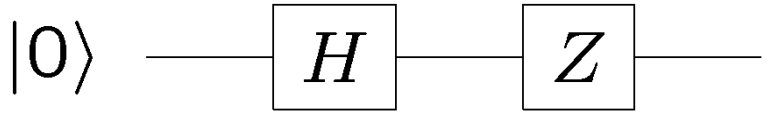


Must design quantum circuits which succeed even when every element of the circuit (preparation, unitary gates, measurements, doing nothing) can fail.

Quantum error correction + Fault-tolerant procedures

Sudden Approximation



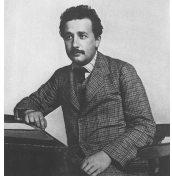


Quantum circuit model

Computing in the sudden approximation

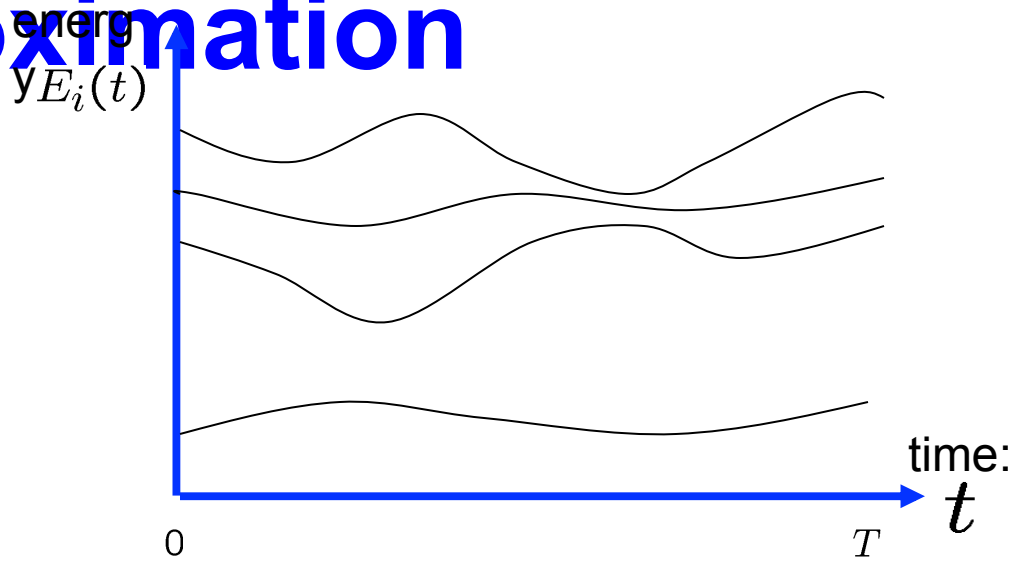
What is the Second Approximation You Learn in Quantum Theory?

Adiabatic Approximation



Time dependent Hamiltonian:

$$H(t)$$



If we prepare $|\psi_0(0)\rangle$, and $H(t)$ varies "slow enough", then at time T the final state will have a large overlap with $|\psi_0(T)\rangle$

"roughly": $T \geq O\left(\frac{1}{E_{gap}^2}\right)$

Adiabatic Quantum Algorithms





E. Farhi, J. Goldstone, S. Gutmann, and M. Sipser, 2000

$$H(s) = (1-s)H_0 + sH_p \qquad 0 \le s \le 1$$

1. Prepare ground state of H_0
2. "Slowly" evolve $H(s)$
3. Measure final state, obtaining answer to problem instance *p*

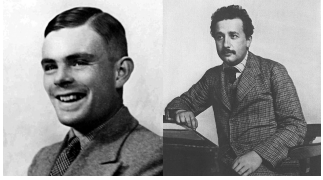
Example: $H_0 = J_z \sum_i (\sigma_x)_i \quad |\psi_0(0)\rangle = \left[\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\right]^{\otimes n}$

$$H_p = \sum_i J_i(p)(\sigma_z)_i + \sum_{<i,j>} J_{i,j}(p)(\sigma_z)_i(\sigma_z)_j$$

Finding ground state of 2D Ising in magnetic field when bond values are problem instance is NP-complete

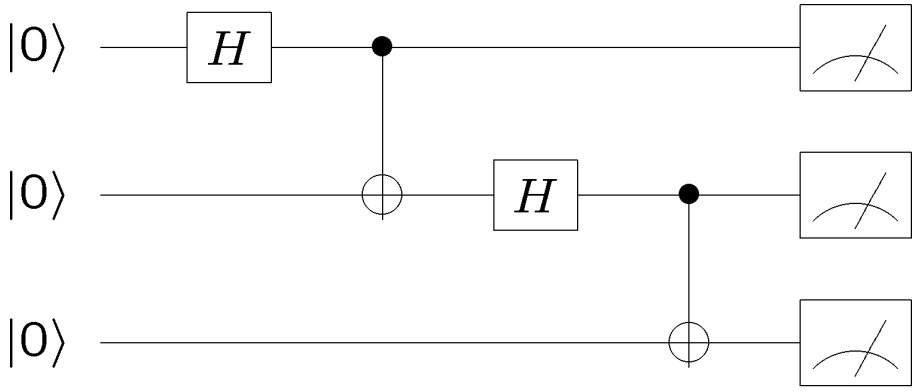
Power of adiabatic quantum algorithms? **Much debate!**

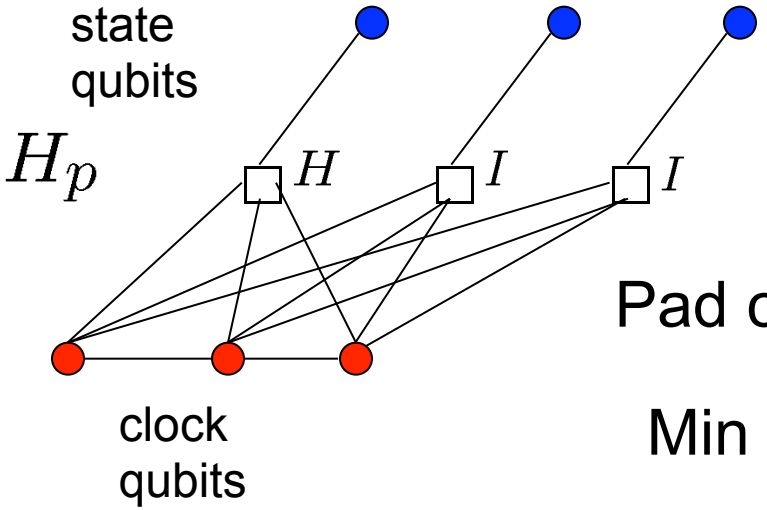
Universal Adiabatic Quantum Computation



Adiabatic quantum computation ⇔ The quantum circuit model

[D. Aharonov, W. van Dam, J. Kempe, Z. Landau, S. Lloyd, O. Regev, 2004]





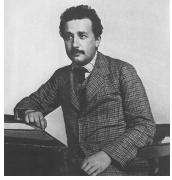
Ground state is history of circuit:

$$|\psi_0\rangle = \frac{1}{\sqrt{s}} \sum_{i=1}^{s} |\psi(i)\rangle \otimes |i\rangle_{clock}$$

Pad circuit with identities and measure

Min gap = O(1/poly(n))

But Realistic?



3-qubit Hamiltonians

[D. Aharonov, W. van Dam, J. Kempe, Z. Landau, S. Lloyd, O. Regev, 2004]

2-qubit Hamiltonians

[J. Kempe, A. Kitaev, O. Regev, 2004]

2-qubit Hamiltonians in a plane

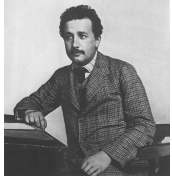
[R. Oliveira, B. Terhal 2005]

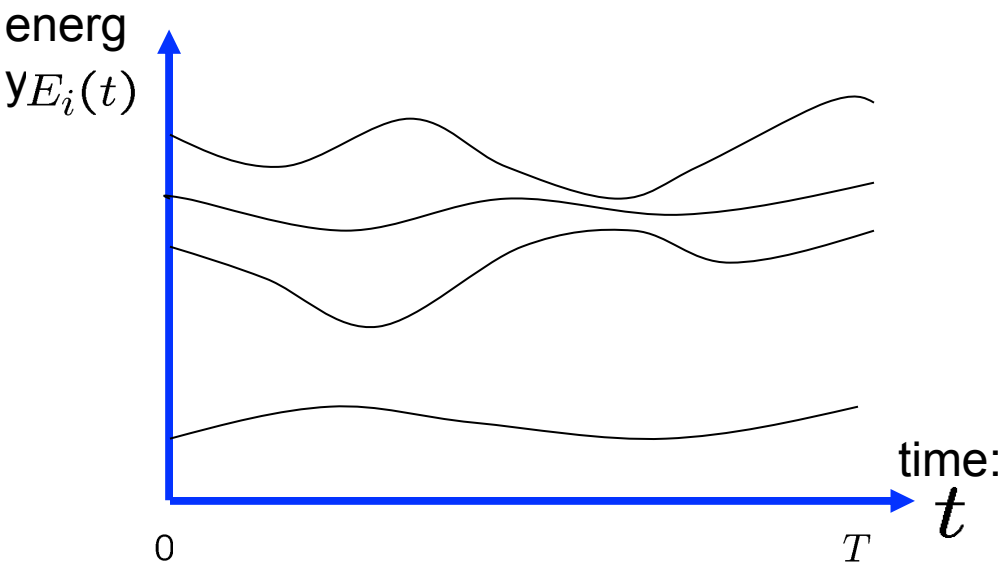
2-body Hamiltonians in a plane with spatial degress of freedom

[A. Mizel, D.A. Lidar, M. Mitchell 2006]

A new **architecture** for building a quantum computer?!?!

But Fault-Tolerant?





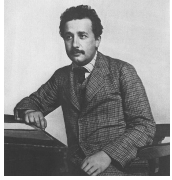
$$E_{min} = O(1/poly(n))$$

Stay in ground state:

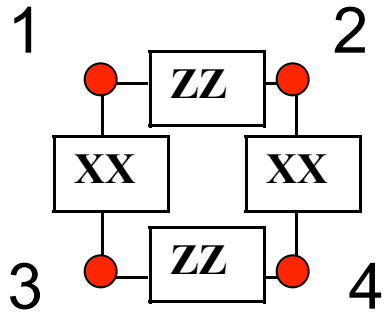
$$kT \ll E_{min}$$

Unreasonable !

A First Step



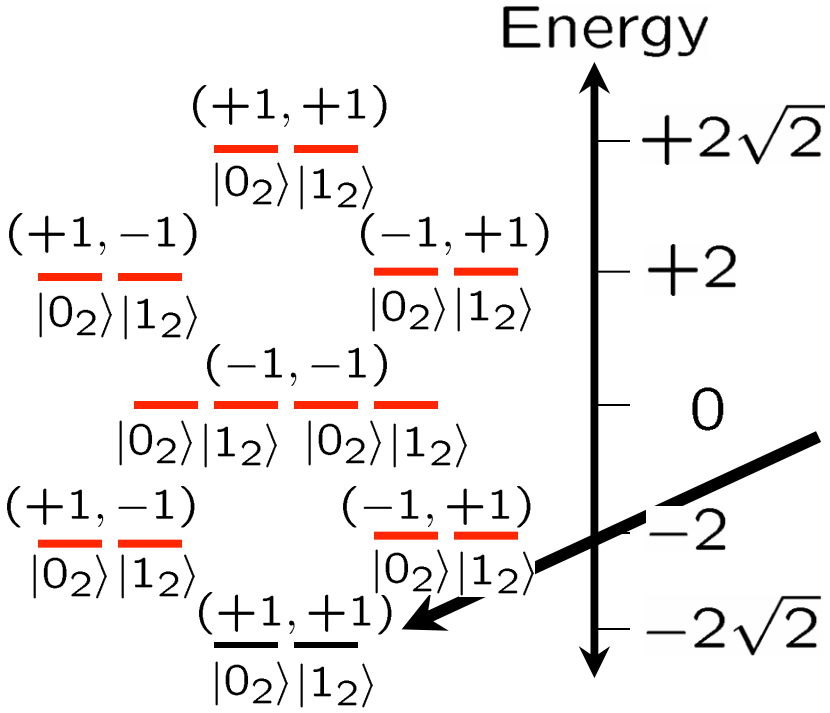
QEC Lesson: it matters what the environment sees!



Hamiltonian

$$H = -(Z_1 Z_2 + Z_3 Z_4 + X_1 X_3 + X_2 X_4)$$

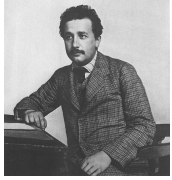
Quantum error detecting code

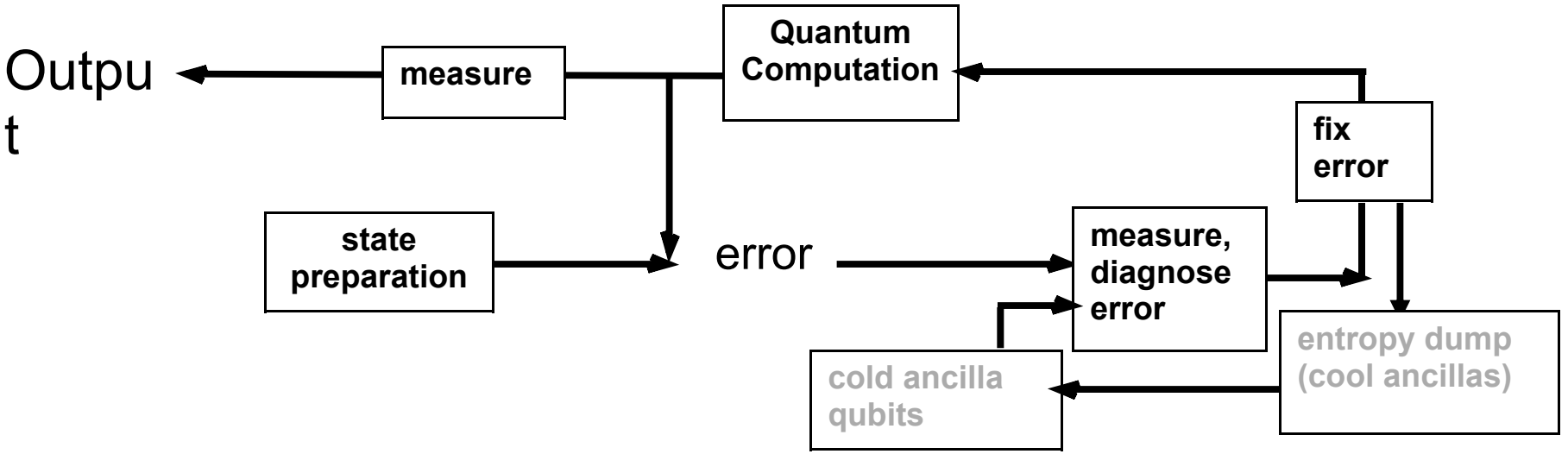


Single qubit couplings see a constant gapped system!

[D. Bacon, K. Brown, K.B. Whaley, Phys. Rev. Lett. 2001]

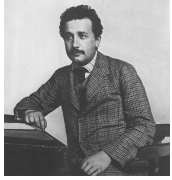
Quantum Error Correcting Is Sophisticated Cooling

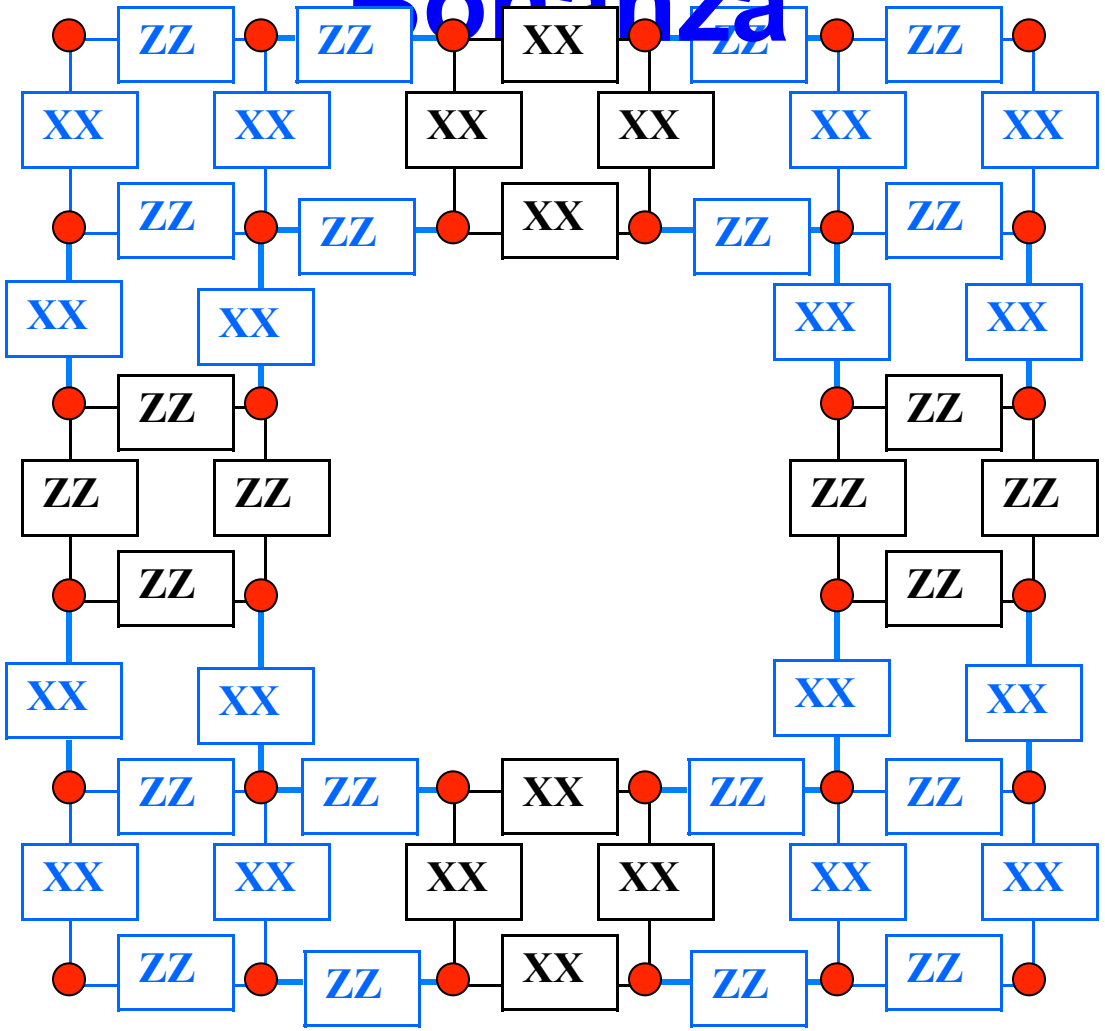


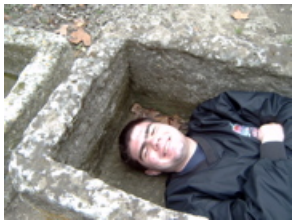


Can we design systems whose eigenstates are quantum error correcting code states & such that keeping the system cool will be equivalent to quantum error correction?

Simulation Bonanza







The Search For a New Phase

The Journey

Symmetry In Shor's Algorithm



Clebsch-Gordan Transform



Information Theory Protocols





Efficient Circuits



Algorithms?



Heisenberg HSP Algorithm



Role of Entanglement in Quantum Computing



New Ideas for Simulating Quantum Systems

What To Simulate?



Quantum Error Correction



Sudden and Adiabatic Approximation



Fault-Tolerant Adiabatic Quantum Comptuers?

Fin



Gregory Crosswhite
Grad Student
UW Physics



Paul Pham
Grad Student
UW Computer Science & Engineering

Thomas Decker
Postdoc
UW Computer Science & Engineering

Work described here in collaboration with:
Andrew Childs (Caltech), Wim van Dam (UCSB), Ike Chuang (MIT)
Aram Harrow (Bristol), Ken Brown (Georgia Tech), Mark Oskin (UW),
Birgitta Whaley (Berkeley)

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